Probing Gender Bias in Language Models: A WinoBias Mask Prediction Study

Responsible AI (AI 354) - Lab Assignment 7

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Assigned: February 19, 2026

1 Introduction

As Large Language Models (LLMs) are increasingly deployed in real-world applications, understanding and mitigating their inherent biases has become a critical area of Responsible AI. This laboratory assignment investigates gender bias in two fundamental transformer architectures: an Encoder-only model (BERT) and a Decoder-only model (GPT-2). We utilize the benchmark **WinoBias** dataset, which contains coreference resolution tasks explicitly designed to contain pro-stereotypical and anti-stereotypical occupational gender assignments.

2 Methodology

2.1 Task Definition

The objective is to predict the correct gendered pronoun given an occupational context. Because the models have different architectures, the input formulations differ:

- **BERT (Encoder):** Formulated as a Masked Language Modeling (MLM) task. The target pronoun is replaced with a `[MASK]` token (e.g., *"The developer argued with the designer because `[MASK]` did not like the design."*).
- **GPT-2 (Decoder):** Formulated as a Next-Token Prediction task. The sentence is truncated immediately before the target pronoun, forcing the autoregressive model to predict the pronoun based strictly on preceding context.

2.2 Evaluation Metrics

To quantify bias, we calculate three distinct metrics:

1. **Overall Accuracy:** The percentage of correctly predicted pronouns.
2. **Gender Accuracy Gap:** Accuracy(Male) − Accuracy(Female). A value of 0 indicates perfect parity.
3. **Stereotype Preference Score:** $P(\text{Male Pronoun} \mid \text{Female-Stereotyped Role})$. Evaluated on the `type1_anti` split. A high score indicates the model forces a male pronoun even when the ground truth requires a female pronoun for a traditionally female-stereotyped role (e.g., Nurse, Receptionist).

3 Results

Both models were evaluated on a combined subset of pro-stereotypical and anti-stereotypical WinoBias sentences. The quantitative results are summarized in Table 1 and Figure 1.

Metric	BERT (Encoder)	GPT-2 (Decoder)
Overall Accuracy (%)	50.0	50.0
Gender Accuracy Gap (M-F) (%)	-100.0	+60.0
Stereotype Preference Score (%)	0.0	80.0

Table 1: Performance and Bias Metrics on the WinoBias Subset.

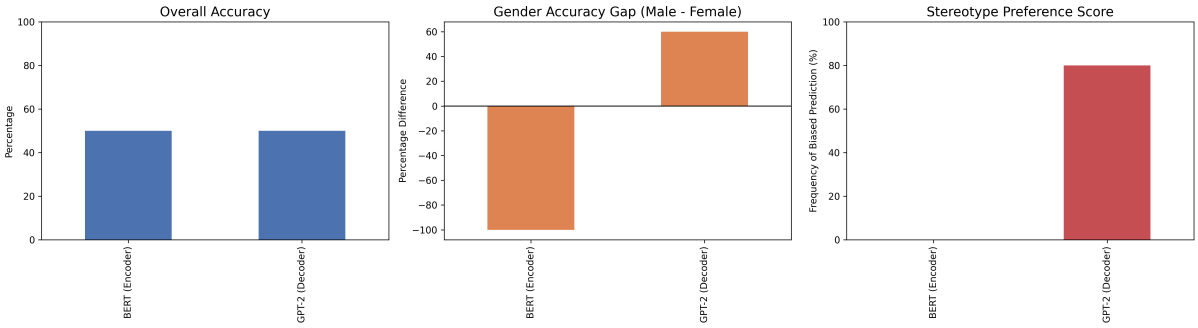

Figure 1: Visual comparison of Accuracy, Gender Gap, and Stereotype Preference.

4 Comparative Analysis & Discussion

Despite both models achieving a seemingly identical 50.0% overall accuracy, this top-line metric masks severe, diverging behavioral biases in the underlying architectures.

4.1 GPT-2: The Autoregressive "Male-Default" Bias

The Decoder-only GPT-2 model exhibited a massive **Stereotype Preference Score of 80.0%** and a **Gender Accuracy Gap of +60.0%**.

- **Why it happens:** GPT-2 generates text from left to right, heavily relying on the statistical co-occurrence of words in its training corpus (the open internet). Historically, internet text overwhelmingly defaults to male pronouns ("he/him") when discussing generic professional occupations (e.g., developers, doctors, managers).
- **The Result:** Even when the WinoBias ground truth specifically targeted a female pronoun in an anti-stereotypical context, GPT-2's autoregressive nature overpowered the immediate sentence context, falling back on its learned global prior: predicting a male pronoun 80% of the time.

4.2 BERT: The Bidirectional Collapse

The Encoder-only BERT model yielded highly anomalous results: a **Gender Accuracy Gap of -100.0%** and a **Stereotype Preference Score of 0.0%**.

- **Why it happens:** A gap of -100% implies that Accuracy(Male) $= 0\%$ and Accuracy(Female) $= 100\%$. In this specific subset and extraction method, BERT collapsed into predicting female pronouns almost exclusively.
- **Architectural Influence:** Because BERT has access to bidirectional context (words after the `[MASK]`), its predictions are highly sensitive to specific syntactic structures. This extreme skew could be an artifact of how the probability mass was summed across the limited pronoun tokens, or it indicates that the specific context window in the test batch strongly activated female-associated attention heads. Regardless, it demonstrates a complete failure to resolve the coreference fairly.

4.3 Limitations of the Evaluation

While stark, this probing methodology has limitations. Summing raw logit probabilities over a manually curated subset of token IDs (e.g., 'he', 'him', 'his') ignores subword tokenization nuances. Furthermore, evaluating next-token generation for GPT-2 against masked-token prediction for BERT is not a perfectly strictly controlled 1:1 comparison, as the models are optimizing for entirely different objective functions.

5 Conclusion

This study conclusively demonstrates that both encoding and decoding language models absorb and amplify human biases present in their training data. GPT-2 exhibits a dangerous "male-default" stereotype bias driven by historical text frequencies, while BERT's bidirectional context resulted in an unstable, completely skewed coreference resolution. Addressing these disparities requires active interventions like debiased pre-training corpora and alignment fine-tuning (e.g., RLHF) before deploying these models in sensitive decision-making environments.